Huautla Mazatec Complex Segments as Single Phonemes: A Multi-layered Subsegmental Structure Approach

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Abstract

This paper proposes a novel approach to the analysis of consonantal phonetic sequences in Huautla Mazatec. These sequences comprise a large inventory of both simple and complex phonemes, including typologically unexpected contrasts such as pre- vs. post-aspirated consonants like /^{h}t/ vs. /t^{h}/ or /^{hn}d/ vs. /^{n}t^{h}/. This paper argues that the primary phonological contrast between these phonemes does not reside in temporal anchoring or linearity, but rather in a distinct arrangement within a subsegmental hierarchical binary structure. This approach adapts Pike & Pike's (1947) 'Immediate Constituents' proposal to the

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[2] Google's Gemini 1.5 Flash AI tool was employed to review and enhance the writing, as my native language is not English. It is essential to clarify that the tool was used for editing purposes only, and no content was generated by the AI.

representational framework of Q-Theory (Garvin et al. 2018; Inkelas & Shih 2016; Shih & Inkelas 2019).

Keywords: Mazatec, consonantal sequences, complex consonants, secondary articulation, pre-aspiration, post-aspiration, Q-Theory, phonological recursion

1. Introduction

The Huautla Mazatec language possesses a large inventory of surface consonantal phonetic sequences formed by combining simple segments and secondary articulation gestures, as observed in (1).

(1) a. [tã͡ʊ̃^{2}][3] ‘money’

b. [tʰo^{1}] ‘rifle’

c. [kˀḛ̃͡ẽ^{2}] ‘dead’

d. [nᵈa^{2}] ‘good’

e. [ntʰa͡ʊ^{1}] ‘wind’

f. [ʰta^{1}] ‘voice’

g. [nʱĩ^{4}] ‘blood’

h. [n̥͡no^{1}] ‘cornfield’

i. [ʔnᵈi^{1}] ‘sterile’

j. [ʃka^{1}] ‘leaf’

[3] Tone notation is as follows: ^{1} low tone, ^{2} mid tone, ^{3} semi-high tone, ^{4} high tone. Sequences of tones represent contour tones.

These sequences can be categorised into four different groups according to their constituents. The first group consists of a consonant followed by a glottal feature (1b, 1c, 1e, 1g). The second group consists of a consonant preceded by a glottal or another glottalised/breathy consonant (1f, 1h). The third group consists of a nasal followed by a plosive or plosive secondary gesture (1d, 1e, 1i). The fourth group consists of a strident fricative-plosive sequence (1j).

As will be treated in §2, the phonologisation of these sequences has been the subject of numerous proposals since Pike & Pike's (1947) classic work[4]. Notably, the distinction between the typologically unexpected (Silverman 2003) pre-aspirated and post-aspirated consonants, especially plosives (1b, 1f), has been the source of the theoretical proposals of Steriade (1994), Goldston & Kehrein[5] (1998), and García García[6] (2013) and Chávez Peón Herrero & Filio García (2021)[7]. The main contention between these works is the treatment of the aforementioned phonetic sequences. One perspective views each phonetic segment as an individual phoneme, while the other perspective views them as the constituents of complex phonemes. Complex phonemes are understood as phonemes the phonetic expression of which involves more articulatory gestures beyond the basic three for consonants (voicing, place, and manner) and vowels (height, backness, and rounding).

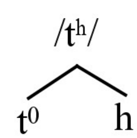
While all the aforementioned proposals treat strident fricative–plosive phonetic sequences (1j) as such in the phonology, there is a divergent phonological treatment for all

[4] P&P from now on.
[5] G&K from now on.
[6] García García (2013) provides the most extensive phonetic description of Huautla Mazatec published to date. While I differ from the phonological analysis presented in that work, I recommend it to readers interested in delving into the phonetic details of this language.
[7] García García (2013) and Chávez Peón Herrero & Filio (2021) will be referred to collectively as CIESAS from now on, as that is the institution of the authors of both works: the Centro de Investigaciones y Estudios Superiores en Antropología Social of Mexico. An initial version of this proposal was originally presented by García García, Chávez Peón, and Polian at the COLOV V conference in 2012.

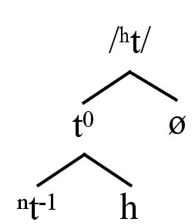
other types of phonetic sequences. For example, P&P (1947) and Steriade (1994) phonologise the contrast of laryngeal timing differences in plosive /t/ in [$t^{h}o^{1}$] 'rifle' and [$^{h}ta^{1}$] 'voice' as merely a distinction between phoneme ordering in the onset: /tho^{1}/ vs. /hta^{1}/. In contrast, G&K attribute it to a difference in the timing of the glottal feature, occurring either within the vowel /to̤^{1}/ or within the consonant /t̤a^{1}/, or both. The CIESAS proposal, on the other hand, adopts a combination of both approaches: /$t^{h}o^{1}$/ vs. /hta^{1}/.

This work proposes a multi-layered arrangement of binary branching structures (2 & 3) to represent the distinction between pre-aspirated /^{h}t/ and a post-aspirated /t^{h}/. I employ a modified version of the notion of subsegmental decomposition, as outlined in Q-theory (Garvin et al. 2018; Inkelas & Shih 2016; Shih & Inkelas 2019) to describe the process that gives rise to the superficial forms of these phonemes. I utilise a parenthetical boundary notation to represent these structures. For example, the structures represented in (2) and (3) are also represented as (t (h)) and (t (Ø)((h))).

(2)



(3)



This proposal will be further elaborated on in §3, where the notion of subsegmental composition of phonemes as outlined in Q-theory (Garvin et al. 2018; Inkelas & Shih 2016; Shih & Inkelas 2019) is introduced to explain the phonological relationship between t^0 and the other dependent subsegments, and to elucidate the derivation steps leading to its surface manifestation. Furthermore, the concept of 'templates,' abstract subsegmental structures composed of variable entities derived from the binary structures, will be proposed to address certain predictions. In §4, conclusions and suggestions for further research will be presented.

2. Phonologisation of Huautla's Consonants Sequences: A Decades-long Problem

Huautla Mazatec is an Otomanguean language (Campbell 2017a; Rensch 1976) spoken by approximately 60,000 people, primarily in the municipalities of Huautla de Jiménez, Santa María Chilchotla, and San José Tenango in the state of Oaxaca, Mexico (Gobierno de México 2024). It is considered the most prestigious of all the Mazatec languages due to the political centrality of Huautla city within the Mazatec Sierra region.

The first linguistic descriptions of Huautla Mazatec come from the early descriptions by Eunice (1937, 1957, 1967) and Kenneth Pike (1948) as part of their SIL's missionary activity. P&P (1947: 81) argue that the claim that sequences containing an obstruent or an affricate and /h/ (e.g., /th/) are not 'single complex aspirated consonants' is supported by the existence of reverse sequences are possible (e.g., /ht/). Conversely, the ability of /h/ and /ʔ/ to occur both before and after sonorants serves as evidence against a complex phoneme analysis for these sequences. The phonologization of phonetic sequences in Huautla Mazatec is illustrated in (4), following the framework proposed by P&P.

(4)	a. [tã͡ʊ̃^{2}]	/tãõ^{2}/	‘money’
	b. [tʰo^{1}]	/tho^{1}/	‘rifle’
	c. [kˀḛ̃͡ẽ^{2}]	/kʔẽ^{2}/	‘dead’
	d. [nᵈa^{2}]	/nta^{2}/	‘good’
	e. [ntʰa͡ʊ^{1}]	/nthao^{1}/	‘wind’
	f. [ʰta^{1}]	/hta^{1}/	‘voice’
	g. [nʱĩ^{4}]	/nhĩ^{4}/	‘blood’
	h. [n̥͡no^{1}]	/hno^{1}/	‘cornfield’
	i. [ʔnᵈi^{1}]	/ʔnti^{1}/	‘sterile’
	j. [ʃka^{1}]	/ʃka^{1}/	‘leaf’

The phonemic inventory of consonants provided by P&P is shown in Table 1. Regarding the native phonemes, it is a rather simple and typologically average system, close to the “maximally ordinary” ideal (Hayes 2009: 31; *cf.* Maddieson 1984); bolded symbols indicate phonemes only found in Spanish loanwords. P&P also propose four vowel phonemes /i e a o/ that can be oral or nasal, and four tone heights; vowels can form sequences up to three elements and clusters of tones of two and three heights can occur both in simple morphemes and inflected forms.

Table 1. P&P (1947) Huautla Mazatec's consonantal phonemes

	Labial	Coronal	Postalveolar	Retroflex	Palatal	Velar	Glottal
Voiceless plosives	**p**	t				k	ʔ
Voiced plosives	**b**						
Voiceless affricates		t͡s	t͡ʃ	t͡ʂ			
Voiceless fricatives		s	ʃ				h

Voiced fricatives	β	**ð**				**ɣ**	
Nasals	m	n			ɲ		
Trills			**r**				
Flaps			ɾ				
Laterals			l				
Approximants					j		

A phonetic rule mentioned by P&P that must be noted regarding consonants is the voicing of /k/ and /t/ after /n/, except when followed by /h/[8] or in a small number of Spanish loanwords. According to P&P, consonants can occur up to three in the syllabic margin position. Importantly, Huautla Mazatec does not allow codas. The analysis of Huautla Mazatec syllables proposed by P&P involves applying successive divisions of layers of 'immediate constituents', analogous to the tagmemic analysis of sentences. Figure 1 exemplifies the proposed division of /ncʔoai^{34}/[9] "you (sing.) notify" ([^{n}t͡s^{ʔ}o̰aḭ^{21}]):

[8] This rule can be extended to the sequences of nasal+affricates and those containing [h].
[9] This representation follows the Mesoamerican notation of tones, where ^{1} the highest and ^{4} the lowest tone (cf. Yip, 2002: 21)

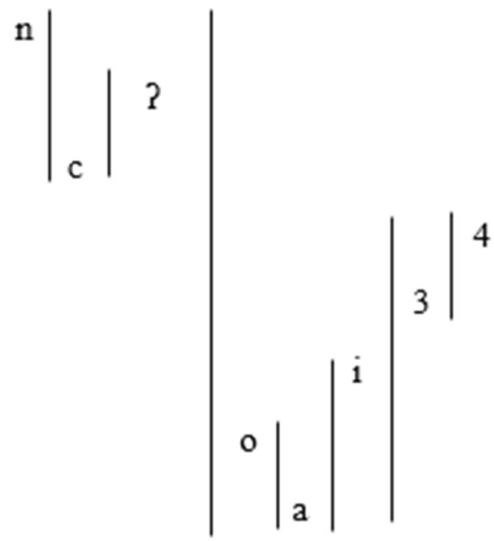


Figure 1. Successive layering /ncʔoai^{34}/ "you (sing.) notify". Reproduced from P&P (1947: 1991)

The proposed successive layering of onsets is based on distributional reasons. P&P observed that in two-consonants clusters, one of the members must belong to a reduced set of the overall consonant inventory, namely /h/, /n/, /ʔ/, /s/, or /ʃ/. These are considered subordinate consonants. While /n/ can only precede stops and affricates (e.g., /nt/, /nt͡s/), and /s/ and /ʃ/ may only precede stops (e.g., /sk/, /ʃt/)[10], /h/ can occur both before and after any other of the non-glottal and non-liquid native consonants, both before and after them (e.g., /hn/, /nh/, /th/, /ht). Similarly, /ʔ/ can occur before sonorants (e.g., /ʔn/) and after both sonorants and obstruents (e.g., /tʔ/, /jʔ/).

The proposed layering of these sequences is strictly linear, dividing the first segment, always subordinate, from the following two: /ʔnt/ → /ʔ/+/nt/, /nkh/ → /n/+/kh/, /h t͡sʔ/ → /h/+/t͡sʔ/, /skʔ/ → /s/+/kʔ/. The layering of the remaining phonemes follows the rules for two-

[10] P&P mention the existence of one word with /ʃn/ sequence, /nka^{3}ʃni$^{3\text{-}4}$/ ([ᵑga^{2}ʃni^{21}]), the native Mazatec name for Chiquihuitlán. However, the actual name for Chiquihuitlán in Huautla Mazatec is [ᵑga^{2}ʃi^{2}ni^{2}], so /ʃn/ should be excluded from any proposed inventory of consonant sequences.

consonant clusters, where /n/ and the glottals are always the subordinate consonants and the plosives are the principal consonants.

While P&P analysis relies on strictly hierarchical and positional arguments, it effectively predicts why stops and affricates are voiced following nasals (/nt/ → [nd], /hnt/ → [hnd]) except when followed by /h/ (/nth/ → [nth]). This voicing difference between [hnd] and [nth] arises from the fact that in the first sequence, the layering /h/+/nt/ vs. /n/+/th/ implies that the voicing rule can only apply when /n/ precedes /t/ in the initial layer.

Steriade (1994: 221) observes that the Mazatec onsets proposed by P&P pose challenges from a sonority perspective, as some violate the ideal organisation of segments within the syllable according to Sonority Sequencing Principle (SSP) (*cf.* Blevins 1995). Steriade (1994) resolves this by proposing that the phonemic status of these Huautla Mazatec sequences is underlyingly that of phonemes sequences, but they surface as complex segments. For example, /nt/ emerges as [^{n}d] and /hnk/ as [$^{h\eta}$g]. The analysis is couched within the framework of 'Aperture Theory', which posits that individual segments possess distinct phonological prosodic positions in which single-valued features can be anchored. These timing slots, called A-slots, are contingent upon each segment's specific stricture and can be classified into three types: A_{max}, present in approximants; A_f, present in fricatives; and A_0, present in unreleased stops. The combination of these A-slots gives rise to other kind of segments: A_0+A_{max} characterises released stops (plosives and nasal stops), and A_0+A_f characterises affricates.

Revisiting P&P's proposal, Steriade (1994: 205) highlights the prominent role of /n/, /h/, and /ʔ/ as 'satellites' by observing that they occur with a wider range of principal consonants than /s/ or /ʃ/, which are limited to combinations with stops and affricates. A key distinction between these two types of sequences lies in their surface realisation: sequences involving /n/, /h/, or /ʔ/ emerge as complex phonemes, while those involving /s/ or /ʃ/ remain as sequences of phonemes. Steriade (1994) explains the ability of /n/, /h/, and /ʔ/ to function as satellites by arguing that these segments represent always-positive, single-valued features with no negative counterparts: [+nasal], [+spread glottis], and [+constricted glottis].[11]

The existence of pre-aspirated, pre-glottalised, post-glottalised, post-aspirated, and plain prenasalised plosives and affricates is explained within Aperture Theory by the fact that plosives and affricates can host a glottal feature and/or a [+nasal] feature in their A_0 and/or A_{max} slots, giving rise to the pre-aspirated, pre-glottalised, post-glottalised, and post-aspirated plosives and affricates series. With the exception of pre-glottalised consonants, all those combinations are found in Huautla Mazatec. Nasals can contrast by being pre-aspirated, pre-glottalised, post-glottalised, post-aspirated, and plain; all those combinations are attested in Huautla Mazatec.

In all these processes, it is assumed that single-valued features belong to independent phonological segments underlyingly, but they anchor to the 'principal' segment after the elision of their time slot anchoring segment and subsequent reassociation of the single-valued feature in the derivation, giving rise to complex segments on the surface. This mechanism is illustrated in Figure 2.

[11] Although single-valued or privative features do not exhibit inherent positive or negative contrasts, I adhere to Steriade's original representation, depicting these features using the [+value] notation.

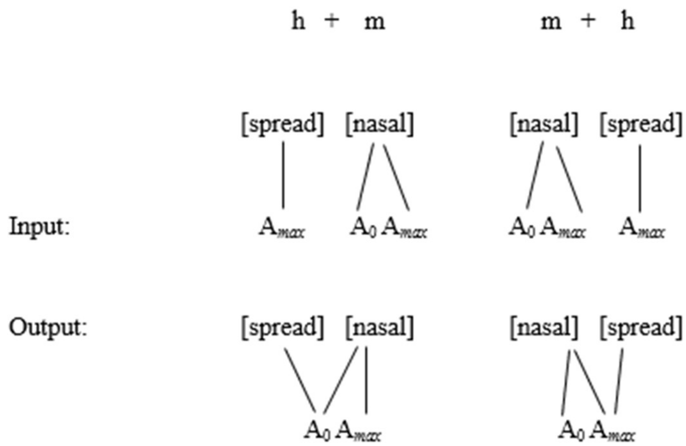


Figure 2. Derivation from /hm/ and /mh/ to [ʰm] and [mʰ] (Reproduced from Steriade (1994: 234)).

According to "Aperture Theory", continuant consonants, such as fricatives and approximants, possess only one timing slot. This implies that continuants cannot anchor more than one single-valued feature nor distinguish the same single-valued feature in two different timing positions. This prediction holds true for fricatives in Huautla Mazatec, as they contrast only as post-aspirated, post-glottalised, or plain. However, it underpredicts the distinction between pre-aspirated and pre-glottalised vs. post-aspirated and post-glottalised approximants[12] and laterals[13].

G&K propose an alternative approach to Huautla Mazatec sequences, arguing that complex phonetic sequences constitute complex phonemes. They posit that the difference between [ʰtV] and [tʰV] does not arise from variations in the phonological segmental anchoring of the glottal feature within /t/ but rather from differences in the anchoring of the

[12] These contrasts are analytically justified by Steriade (1994) on the basis that the sequences [hβ] and [βh] are phonologically /hb/ and /bh/, respectively, thus supporting the prediction that obstruents can host glottal features in either of their two-timing slots, while the sequences [ʔj] and [ʔβ] map onto /ʔi/ and /ʔo/ respectively.
[13] Steriade (1994: 240, 243) considers that /lʔ/ sequences are possible because /l/, a lateral approximant, is a continuant and thus count with an A_{max} slot, which express the glottal single-valued feature to the right and forbids the existence of /ʔl/ sequences. However, in Huautla Mazatec this contrast does exist, as can be seen in the words [le2ˈl̰͡le1] "listless" vs. [ˈlḭ͡i] "fire" (Mejía Carrera et al. 2022: 364).

feature within the syllable. Specifically, in [ʰtV], the [spread glottis] feature is anchored to the onset, i.e.: /t̤V/, whereas in [tʰV], the feature is anchored to the vowel, i.e.: /tV̤/.

This model suggests that the 'satellites' proposed by Steriade (1994) are autosegment-like privative features, e.g., [nasal], [spread glottis], and [constricted glottis], that associate with both consonants and vowels (Figure 3). G&K support their proposal by pointing out strong OCP-like restrictions against the co-occurrence of consonants and vowels marked with the same of any of these features.

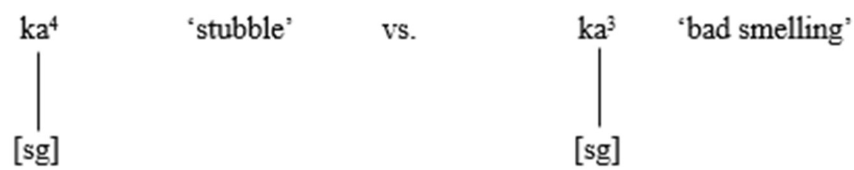


Figure 3. The underlying difference between surface [ʰka^{4}] 'stubble' and [kʰa^{3}] 'bad smelling' (Reproduced from G&K (1998: 313)).

This representation implies that there are no distinctions of timing anchoring of glottal features at the phonological level. Consequently, plosives and affricates can be plain, pre-nasalised, aspirated, glottalised, pre-nasalised aspirated, or pre-nasalised glottalised, while fricatives, nasals, and approximants can be plain, aspirated, or glottalised. Vowels can be oral or nasal, and creaky or breathy. The model predicts that the features will always be expressed to the left of the host segment in the phonological level. This prediction is indeed attested in the data shown in (4).

(4)	a. [tã͡õ^{2}]	/tãõ^{2}/	'money'
	b. [tʰo^{1}]	/to̤^{1}/	'rifle'
	c. [kˀḛ̃͡ẽ^{2}]	/kḛ̃^{2}/	'dead'

d. [n^{d}a^{2}]	/^{n}ta^{2}/	‘good’
e. [nt^{h}a͡ʊ^{1}]	/^{n}ta̤o^{1}/	‘wind’
f. [^{h}ta^{1}]	/t̤a^{1}/	‘voice’
g. [n^{ɦ}ĩ^{4}]	/ni̤^{4}/	‘blood’
h. [n̥͡no^{1}]	/n̤o^{1}/	‘cornfield’
i. [ʔn^{d}i^{1}]	/^{n}t̰i^{1}/	‘sterile’
j. [ʃka^{1}]	/ʃka^{1}/	‘leaf’

Following Kingston’s (1985) Binding Principle, the CIESAS proposals posit the existence of phonological aspirated obstruents that express the glottalic feature to the right, encompassing both plain (5b) and pre-nasalised variants (5e). Furthermore, they propose glottalised sonorants and aspirated sonorants, typically expressing the glottalic feature to the left (5h). Post-glottalised obstruents, consistent with the proposal of G&K, are analysed as having the glottal feature associated with the vowel (5c). Conversely, pre-aspirated plosives, both plain (5f) and prenasalised, are analysed as sequences of a phonological /h/ followed by an obstruent that exhibits pre-aspiration. This analysis is supported by the argument that /h/ in this context behaves distributionally like the other two fricatives of the system, /s/ and /ʃ/.

(5)	a. [tã͡ʊ̃^{2}]	/tãõ^{2}/	‘money’
	b. [t^{h}o^{1}]	/t^{h}o^{1}/	‘rifle’
	c. [k^{ˀ}ḛ̃͡ẽ^{2}]	/kḛ̃^{2}/	‘dead’
	d. [n^{d}a^{2}]	/^{n}ta^{2}/	‘good’
	e. [nt^{h}a͡ʊ^{1}]	/^{n}t^{h}ao^{1}/	‘wind’
	f. [^{h}ta^{1}]	/hta^{1}/	‘voice’

g. [n^{ɦ}ĩ^{4}]	/n̤i^{4}/	‘blood’
h. [n̥͡no^{1}]	/n̤o^{1}/	‘cornfield’
i. [ʔn^{d}i^{1}]	/^{n}t̰i^{1}/	‘sterile’
j. [ʃka^{1}]	/ʃka^{1}/	‘leaf’

This proposal adopts G&K assumption that post-glottalised consonants arise from vowel creakiness, a phenomenon observed in Huautla Mazatec evidence. While García García (2013) and Chávez Peón & Filio (2021) do not consider vowel breathiness, as it is unattested in Huautla Mazatec, they attempt to address the distinction between pre-aspirated and post-aspirated sonorants by treating it as an allophonic variation within aspirated sonorants, allowing them to be both pre-aspirated and post-aspirated (5g, 5h).

While some speakers exhibit a tendency to neutralise [nh]-like sequences to [hn], this clearly represents a historical phonemic merger. However, this merger does not adequately explain why non-merging speakers still phonologically distinguish between these two sequences. Words that display post-aspirated sonorants, such as the word for ‘blood’, can be pronounced with either post-aspiration [n^{h}ĩ^{4}] or pre-aspiration [n̥͡nĩ^{4}] among merging speakers. However, words like ‘cornfield’ consistently exhibit pre-aspiration [n̥͡no^{1}] and never post-aspiration.

In the next section, I will present a new proposal that incorporates some of the assumptions presented in the previously discussed models; specifically, the hierarchical nature of principal versus subordinate consonants from P&P (1947), the SSP-respecting syllable structure proposed by Steriade (1994), and the existence of complex consonants as suggested by G&K (1998) and CIESAS’ proposals.

3. A Multi-layered Approach: The Case of Huautla Mazatec Consonant System

This work proposes that Huautla Mazatec allows only two types of onsets: C(V) and CC(V). The latter type is restricted to sequences of strident fricatives followed by plosives or affricates. Furthermore, it is proposed that Huautla Mazatec possesses 85 phonemic consonants, as presented in Table 2.

Table 2. Huautla Mazatec Consonants

		Labial	Alveolar	Post-alveolar	Retroflex	Palatal	Velar	Glottal
Plosives	Plain	p	t				k	ʔ
	Pre-aspirated		ʰt				ʰk	
	Post-aspirated		tʰ				kʰ	
	Glottalised		tˀ				kˀ	
	Pre-nasalised	ᵐp	ⁿt				ᵑk	
	Pre-aspirated pre-nasalised		ʰⁿt				ʰᵑk	
	Pre-glottalised pre-nasalised	ˀᵐp	ˀⁿt				ˀᵑk	
	Post-aspirated pre-nasalised		ⁿtʰ				ᵑkʰ	
	Post-glottalised pre-nasalised		ⁿtˀ				ᵑkˀ	
Affricates	Plain		t͡s	t͡ʃ	ʈ͡ʂ			
	Pre-aspirated		ʰt͡s	ʰt͡ʃ	ʰʈ͡ʂ			
	Post-aspirated		t͡sʰ	t͡ʃʰ	ʈ͡ʂʰ			
	Glottalised		t͡sˀ	t͡ʃˀ	ʈ͡ʂˀ			
	Pre-nasalised		ⁿt͡s	ⁿt͡ʃ	ⁿʈ͡ʂ			
	Pre-aspirated pre-nasalised		ʰⁿt͡s	ʰⁿt͡ʃ	ʰⁿʈ͡ʂ			
	Pre-glottalised pre-nasalised		ˀⁿt͡s	ˀⁿt͡ʃ	ˀⁿʈ͡ʂ			
	Post-aspirated pre-nasalised		ⁿt͡sʰ	ⁿt͡ʃʰ	ⁿʈ͡ʂʰ			
	Post-glottalised pre-nasalised		ⁿt͡sˀ	ⁿt͡ʃˀ	ⁿʈ͡ʂˀ			
Fricatives	Plain		s		ʂ			h
	Aspirated		sʰ		ʂʰ			
	Glottalised		sˀ		ʂˀ			
Nasals	Plain	m	n			ɲ		
	Pre-aspirated	ʰm	ʰn			ʰɲ		
	Post-aspirated	mʰ	nʰ					
	Pre-glottalised	ˀm	ˀn			ˀɲ		
	Post-glottalised	mˀ	nˀ			ɲˀ		
Laterals	Plain		l					
	Pre-glottalised		ˀl					
	Post-glottalised		lˀ					
Trills			r					
Taps			ɾ					
Approximants	Plain	β				j		
	Pre-aspirated	ʰβ				ʰj		
	Post-aspirated	βʰ				jʰ		
	Pre-glottalised	ˀβ				ˀj		
	Post-glottalised	βˀ				jˀ		

Plain, pre-aspirated, and pre-glottalised stops and affricates are typically voiced on the surface ($/^{n}t/ \rightarrow [^{n}d]$), although free variation of voiceless realisations also occur ($/^{n}t/ \rightarrow [^{n}t]$) (cf. Herrera Zendejas 2003). It is important to note that voicing does not constitute an active parameter of distinction among Huautla Mazatec obstruents, a fact that will be crucial for the arguments presented in this section.

The delimitation of the concept of phoneme used in this analysis adheres to Trubetzkoy's criteria for monophonematic sequences, as outlined in his *Grundzüge der Phonologie* (Trubetzkoy 1934). Rules I, II, and V will be treated in the following paragraphs to support this proposal.

Rule I pertains to the impossibility of segmenting constituents across syllabic boundaries. As mentioned previously, Huautla Mazatec does not permit codas. This observation is supported by the fact that words in this language (cf. Pike 1947) never exhibit word-final consonants, a strong indicator of coda prohibition.

Rule II delineates the permissible phonetic nature of monophonematic sequences. Trubetzkoy (1939) posits that only sequences exhibiting a single articulatory movement or a progressive release of an articulatory complex can be considered monophonematic. Pre-nasalised stops and affricates, sharing the same place and voicing and differing only in manner of articulation, exemplify a single articulatory movement. Laryngealisation and breathy phonation in pre-nasalised obstruents and sonorants can be interpreted as an integral part of the segment, as it does not disrupt the articulation or the perception of voicing or voicelessness. This criterion excludes sequences comprising a fricative and an occlusive or an affricate with different places of articulation features.

The set of consonants produced through the progressive release of an articulatory complex encompasses affricates and all aspirated and glottalised obstruents. This criterion similarly excludes sequences involving fricatives and occlusives or affricates with the same place of articulation. Complex consonants such as /ⁿt͡sʰ/ exemplify the intersection of the two permissible phonetic realisations of monophonematic sequences.

The notion of intersecting sets of consonants supports Rule V proposed by Trubetzkoy, which posits that monophonematic sequences exhibit symmetry within the phonological system. This symmetry is validated through the establishment of intersecting series of correlations. These correlations involve phonemes whose phonetic expressions share certain characteristics and intersect with another equivalent set.

As can be seen in Table 2, Huautla Mazatec exhibits a rich consonant inventory characterised by an orderly array of intersecting contrasts among its members. For example, the plosives /k/ and /t/ contrast with /ⁿk/ and /ⁿt/ in terms of pre-nasalisation, and with /kʰ/ and /tʰ/ in terms of post-aspiration. Similarly, plain pre-nasalised plosives /ⁿk/ and /ⁿt/ differ from /ⁿkʰ/ and /ⁿtʰ/ in terms of aspiration, and these consonants in turn differ from /kʰ/ and /tʰ/ in terms of pre-nasalisation, and so on.

The accounts of P&P (1947) and Steriade (1994) avoid postulating an inventory of consonants that is unusually large by typological standards (Cf. Maddieson 2005). They achieve this by focusing on explanatory mechanisms grounded in syllable structure. In contrast, G&K observe that complex phonetic sequences occur within monomorphemic words, such as [ntʰa͡ʊ¹] 'wind' and [ʔnᵈi¹] 'sterile.' Since these onsets are not amenable to segmentation by morphophonological or morphological processes, G&K advocate for an analysis that posits complex phonemes. In the present account I concur with this perspective.

On the other hand, G&K's proposal highlights another typological exceptionality of Huautla Mazatec: the presence of pre-aspirated consonants in word-initial position, a feature not attested in any other language (Kehrein & Golston 2004; Silverman 2003)[14]. By proposing that the contrasting position of glottal features depends on a generalised expression of secondary features to the left of segments, and by including vowels as potential hosts for these secondary features, G&K not only address the unusual occurrence of pre-aspirated plosives in Huautla Mazatec but also avoid the need for a phonological account that relies on contrasting timing of glottal features within the language consonants.

However, in this work I maintain that glottal features are exclusively associated with consonantal representations. This perspective necessitates a reconsideration of how to phonologically represent the surface timing contrast observed between [^{h}t] vs. [t^{h}], or [^{n}t^{h}] vs. [^{hn}d].

The recent works of Lionnet (2017) and the Q-theory framework developed by Inkelas et al. (2018; 2016; 2019) highlights the need to reconsider how certain phonetic characteristics of surface forms are encoded in the phonological grammar. Specifically, these approaches advocate for the incorporation of scalar sub-phonemic or subsegmental representations.

Q-theory (Garvin et al. 2018; Inkelas & Shih 2016; Shih & Inkelas 2019) allows for the representation of complex segments by deconstructing them into subsegments, entities that incorporate the phonetic phasing of the surface expression of phonemes (onset, target, and offset), an idea that draws direct inspiration from Steriade's (1994) Aperture Theory.

[14] Bartholomew (1960), Guerrero Galván (2013), and Palancar (2013) propose that Proto-Otomi, another Otomanguean language, possessed pre-aspirated consonants in initial position. While the primary evidence for this comes from diachronic reconstruction, Bartholomew (1960: 317) reports, based on personal communication with Jenkins and Voigtlander, that pre-aspiration was present in San Gregorio Otomí. However, no published works currently corroborate this assertion.

For example, the distinction between /^{h}t/ and /t^{h}/ in Huautla Mazatec could be subsegmentally represented as (h t t) versus (t t h), not only capturing the timing difference between the glottal feature but also highlighting the relative articulatory strength of the plosive phase compared to the glottal phase.

Within the Q-theory framework, the difference between surface forms [^{n}d], [^{hn}d], and [^{n}t^{h}] could be represented subsegmentally as (n d d), (h n d), and (n t h), respectively, incorporating voicing into the phonological representation. However, given that voicing is considered predictable in all descriptions of Huautla Mazatec (cf. Vielma Hernández 2017), it might be considered redundant to explicitly mark it in the underlying representation.

An alternative approach would involve proposing a rule that governs the surface realisation of /^{n}t/ (n t t) and /^{hn}t/ (h n t) as voiced, necessitating the introduction of subsegmental rules. A simple rule could posit that in Huautla Mazatec, when a nasal subsegment precedes an obstruent subsegment, it induces voicing in all subsequent obstruent subsegments within the same phoneme, resulting in (n d d) from (n t t) and (h n d) from (h n t). However, any rule that aims to explain this phenomenon must also account for the lack of voicing in the surface expression of /^{n}t^{h}/ (n t h). The rule mentioned above would require an exception to account for the absence of voicing when a glottal subsegment follows the nasal subsegment, as previously noted by P&P (1947).

Alternatively, within the Q-theory framework, we could incorporate the 'Immediate Constituent' distinction between principal and subordinate consonants proposed by P&P (1947) at the subsegmental level. In this approach, /^{n}t/ would be represented as (n / t t), /^{hn}t/ as ((h) (n / t)), and /^{n}t^{h}/ as ((n) (t / h)). The subsegmental voicing rule would then be applied only when a subordinate nasal subsegment precedes a principal obstruent subsegment within the same layer; typologically, voiced stops are more common after nasal consonants compared to post-nasal voiceless stops (Pater 2015: 272).

While this solution is satisfactory, it raises two important questions. Firstly, why does the subsegmental voicing rule only apply within the same subsegmental layer? Secondly, is there a potentially recursive subsegmental structure within these phonemes, as these representations appear to suggest?

The first question can be addressed by appealing to the inherent limitations of the model itself. The subsegmental voicing rule is explicitly defined to operate within the boundaries of a single subsegmental layer. However, it's important to remember that obstruent voicing in Huautla Mazatec is predictable. Consequently, obstruents are not inherently specified for voicing in the underlying representation. The emergence of voiceless obstruents on the surface is therefore governed by a separate subsegmental rule, similar to the one that governs the realisation of prenasalised, pre-glottalised, and pre-aspirated pre-nasalised obstruents.

While an Elsewhere-like rule could account for the voicelessness observed in plain obstruents, as well as in post-glottalised and post-aspirated plain and pre-nasalised obstruents, I propose that the underlying cause of voicelessness differs in each case. Voicelessness in plain obstruents likely reflects a default state, in accordance with cross-linguistic observations (cf. Archangeli & Pulleyblank 1994: 170; Kenstowicz 1994: 62). In contrast, voicelessness in post-glottalised and post-aspirated plain and pre-nasalised obstruents may be attributed to a general preference for voicelessness in ejective and aspirated obstruents.

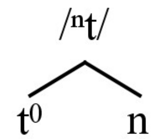
Our primary aim is to demonstrate that the voicing of subsegmental obstruents accompanied by a secondary articulatory subsegment is determined by that secondary subsegment itself. In plain, pre-glottalised, and pre-aspirated pre-nasalised consonants, for example, the nasal subsegment (/n/) induces voicing in the surface realisation of /^{n}t/. Conversely, in post-aspirated and post-glottalised obstruents, the glottal subsegments (h) or (ʔ) are responsible for the resulting surface voicelessness of /t^{ʔ}/ and /t^{h}/.

Voicing is not the only predictable characteristic in complex obstruents; the temporal anchoring of nasal and glottal features within the segmental structure is also predictable. Glottal features typically occur to the right of obstruents in onset position (and vice versa in codas), as observed by Kingston (1985, 1990). Conversely, nasality typically precedes the obstruent, as noted by Riel & Cohn (2011).

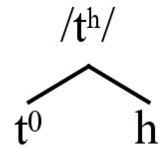
Let us set aside Q-theory representations for now and explore a different route for the phonologisation of Huautla Mazatec complex phonemes. I will start by asserting that the relationship between the secondary subsegment (n) and the principal subsegment (t) in plain pre-nasalised obstruents is the same as the one between secondary glottal subsegments (h) or (ʔ) and the principal subsegment (t) for post-aspirated and post-glottalised obstruents, that is, a relationship of dependency.

Drawing inspiration from early X-bar syntactic approaches (cf. Carnie 2013; Chomsky 1970) and the hierarchical structure between phonological constituents characteristic of Dependency Phonology (Anderson & Ewen 1987), particularly the work of van de Weijer (1996, 2011) on the phonological structure of complex segments, I represent the relationship between the features of the principal and subordinate subsegments using a strict binary structure. In this structure, any secondary feature occupies the dependent branch on the right, while all the features of the main subsegment, represented as X^0, are located on the head branch to the left. The representations of /^{n}t/ and /t^{h}/ are shown in 6 and 7, respectively.

(6)



(7)



Surface realisations of these structures are mediated by a merge process[15] between X^0 and its dependent, during which four phonological rules are applied. To illustrate the derivation of the phonetic manifestations from the underlying representations shown in 6 and 7, I will return to Q-theory. To facilitate this analysis, I will translate these representations into a parenthetical embedding notation. (6) represents $/^{n}t/$ and is expressed as (t (n)), while 7 represents $/t^h/$, is expressed as (t (h)).

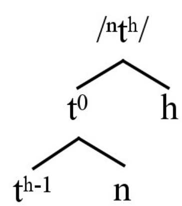
The first rule assigns two positions to the head and one to the dependent, resulting in (t t (n)) from (t (n)) and (t t (h)) from (t (h)). The second rule reorders the dependent subsegment. Nasal subsegments move to the left edge ((t t (n)) → ((n) t t)), while glottal subsegments move to the left edge if the head is a sonorant and remain in the right edge if the head is an obstruent, following Kingston's binding principle ((t t (h)) → (t t (h))). The third rule assigns voice specifications to obstruent phonemes. Nasal subsegments induce

[15] Nasukawa (2017) proposes that merge as an operation isn't exclusive to the FLN component of the language faculty (*contra* Hauser et al., 2002), but also forms part of the FLB module, occurring in processes of internal computation of phonological categories, prior to their externalisation.

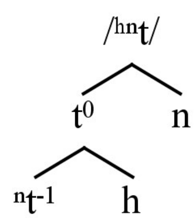
voicing (((n) $t_{[0voice]}$ $t_{[0voice]}$) → ((n) $d_{[+voice]}$ $d_{[+voice]}$)), while glottal subsegments induce voicelessness (($t_{[0voice]}$ $t_{[0voice]}$ (h)) → ($t_{[-voice]}$ $t_{[-voice]}$ (h))). The fourth and final rule deletes the inner parenthetical boundaries, yielding the surface forms of the complex phonemes: ((n) d d) → (n d d) and (t t (h)) → (t t h).

These structures enable us to represent the apparent recursion observed in P&P's Immediate Constituents-inspired Q-theory representations of /$^{n}t^{h}$/ and /^{hn}t/ as ((n) (t / h)) and ((h) (n / t)), respectively. When multiple subordinate subsegments are present, the non-immediate subsegment is represented in a lower layer, with the head position, X^{-1}, occupied by the result of the merge between the principal subsegment X^{0} and the secondary subsegment, which resides in the sister position (8 and 9).

(8)



(9)



Starting with the derivation of /$^{n}t^{h}$/, I translate the representation shown in 8 into a Q-theory parenthetical notation: (t (h) ((n))). At this point, it is crucial to distinguish that the first rule, which fills the subsegmental positions, applies to the entire phoneme. In contrast, the rules governing the timing, and voicing of the obstruent, and deletion of parenthetical boundaries apply only at the boundaries of adjacent subsegments separated by a single parenthetical boundary.

Thus, the first step in the derivation of (t (h) ((n))), which involves assigning two positions to the head subsegment, is inapplicable, as the phoneme already has its three subsegmental positions filled. The second rule pertains to the placement of the secondary subsegment within the first layer, so subsegment (h) remains to the right of the main subsegment. The third rule specifies (t) as voiceless: ($t_{[0voice]}$ (h) ((n))) → ($t_{[-voice]}$ (h) ((n))). The final rule deletes all inner parenthetical boundaries. Since rules can only apply one at a time, the parenthetical boundaries of (h) are deleted, as are the outer boundaries of ((n)), but not the inner boundaries: (t h (n)).

($t_{[-voice]}$ h (n)) corresponds to the subsegment t^{h-1} in (8). A new phase or cycle of rules must be posited to obtain the surface form of /$^{n}t^{h}$/. The first rule in this new merge process involves the placement of (n) with respect to the subsegmental sequence, moving it outside its parenthetical boundaries to the left edge of the (t h) head: ($t_{[-voice]}$ h (n)) → ((n) $t_{[-voice]}$ h). The second rule is inapplicable, as the specification of the (t) subsegment is already in place. The third and final rule deletes the inner boundaries, resulting in the superficial manifestation of /$^{n}t^{h}$/: ((n) $t_{[-voice]}$ h) → (n t h).

The representation of /^{hn}t/ in (9) is translated into Q-theory notation as (t (n) ((h))). Since the three subsegmental positions are already filled, the first step in the derivation involves placing (n) to the left of the plosive subsegment: (t (n) ((h))) → ((n) t ((h))). The second rule specifies the voicing of the plosive subsegment: ((n) $t_{[0voice]}$ ((h))) → ((n) $d_{[+voice]}$

((h))). The final step involves deleting the first layer of inner parenthetical boundaries: ((n) $d_{[+voice]}$ ((h))) → (n d (h)).

The second phase of derivation presents an intriguing challenge to the proposals presented thus far. The first rule, which governs the timing of the dependent subsegment (h) relative to (n d), could theoretically result in either movement to the left edge, as (n) is a sonorant, yielding ((h) n d) according to KBP, or remain at the right edge, as (d) is an obstruent, resulting in (n d (h)). The attested surface form is [^{hn}d], suggesting that the timing anchoring rule for (h) preferentially targets the left sonorant edge.

While this analysis is satisfactory, I do not adopt it here. Instead, I propose that the rule governing the time anchoring of glottal subsegments within projections headed by X^{-1} adheres to an anti-KBP rule: glottal subsegments anchor to the left edge when the X^0 head is an obstruent and to the right edge when the X^0 head is a sonorant[16]. Crucially, the calculus to determine glottal timing considers the X^0 head in any layer, while its domain of application remains within the edges of the projection headed by any X^n. Although ad-hoc, this assumption is crucial for accounting for pre-aspiration in obstruents and post-glottalisation/post-aspiration in sonorants.

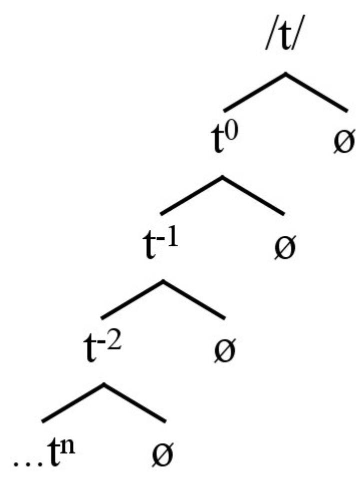
The derivation of (n d (h)) proceeds as follows: the subsegment (h) moves to the left, yielding ((h) n d). Since the subsegment (d) already possesses a voice specification, (h)

[16] While the application of this rule may appear highly ad hoc, the placement of glottal features to the right of sonorants and to the left of obstruents is predicted by KBP for consonants in coda position. This principle suggests that timing tends to be associated with prosodic positions, an idea extensively explored in Kehrein & Golston (2004). Prosodically conditioned alternations of pre-aspiration and post-aspiration of obstruents are observed in languages such as P'urhepecha (Chamoreau 2009), Scottish Gaelic (Iosad 2020), and Faroese (Voeltzel 2022). Although the effects of the so-called "anti-KBP" rule are evident in other languages as a prosodically conditioned consequence of KBP, the unique nature of Huautla Mazatec pre-aspiration can be explained as a non-conditioned, specialized, and thus phonologized, effect of the KBP mirroring phenomenon. This could be an example of a sub-phonemic 'crazy rule' of the III type, as proposed by Chabot (2021: 88). It represents a 'crazy structural change' in the sense that it is usually expected for initial obstruents and sonorants to display a secondary glottalic articulation to the right and left, respectively, and not vice versa.

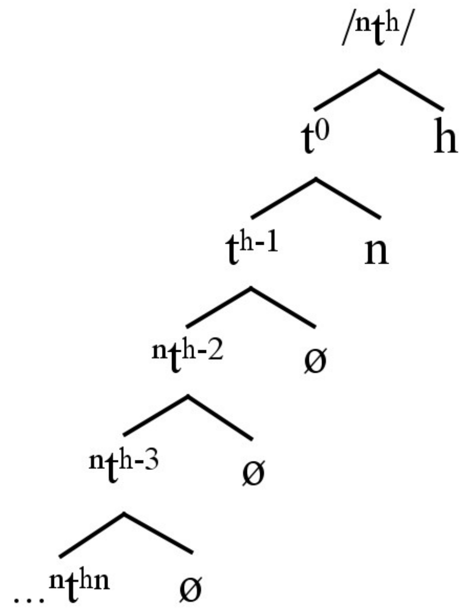
cannot induce voicelessness. Finally, the inner parenthetical boundaries are erased, resulting in the attested pre-nasalised pre-aspirated articulation: (h n d).

Thus far, I have examined the binary representations of complex consonants. However, it is pertinent to inquire whether simple consonants also project the same type of structures. The answer is affirmative. All phonemes possess the capacity to project any number of binary layers. Nevertheless, when no subsegment is specified in the dependent branch, merge remains inactive, and subsequent layers exhibit an identical head, distinguished solely by the representation of decreasing negative numbers: X^0, X^{-1}, X^{-2}, ..., X^n. Void dependent layers are denoted by the symbol Ø, a purely formal theoretical notation. This is illustrated in 10 and 11 using the phonemes /t/ and /$^{n}t^{h}$/ as examples.

(10)



(11)

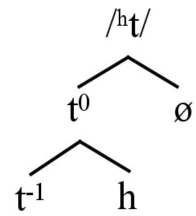


As observed in 11, while the merge processes result in distinct heads for $/^{n}t^{h}/$ in the initial three layers due to the presence of subsegments in the dependent branches of the first two layers, subsequent n-layers merely reiterate the projection of the immediately preceding layer ad infinitum. This phenomenon is even more pronounced in 10, where /t/, lacking any subsegmentally specified dependent branches, exhibits a redundant representation through iterative recursion from the beginning.

The primary objective of these representations is not to demonstrate the theoretically infinite projection potential of Huautla Mazatec phonemes. Instead, their purpose is to highlight the potential existence of void layers. This concept is crucial for explaining the phonological distinction between pre-aspirated and post-aspirated plosives.

$/t^{h}/$ is represented as (t (h)), with the subsegment (h) occupying the dependent branch and (t) serving as the head. In contrast, I propose that the structure of $/^{h}t/$ is (t (Ø) ((h))), as illustrated in 12.

(12)



The structure of /^{h}t/ is supported by three key aspects of the proposed framework: the capacity of phonemes to project multiple layers, the ability of head subsegments to project themselves into lower layers, and the possibility of dependent branches remaining void, that is, unspecified for any subsegment.

The derivation of (t (Ø) ((h))) into (h t t) proceeds as follows. The first rule mandates the filling of all three subsegmental positions with the head. Consequently, as Ø represents a null position, the subsegment (t) occupies the vacant position, resulting in (t t (Ø) ((h))). The first two rules of the initial cycle are applied vacuously: Ø cannot move to the edge, nor can it influence the voice specification of the obstruent subsegment. The third rule not only erases the inner parenthetical boundaries but also eliminates the presence of bare Ø, yielding (t t (h)).

The first step of the second cycle of rules involves the application of the anti-KBP rule. Since (t) occupies the X^0 position, the subsegment (h) moves to the left edge: ((h) t t). The second rule specifies the subsegment (t) as voiceless due to the glottal status of (h): ((h) $t_{[0voice]}$ $t_{[0voice]}$) → ((h) $t_{[-voice]}$ $t_{[-voice]}$). The third and final rule erases the inner boundaries, yielding the surface manifestation: ((h) t t) → (h t t).

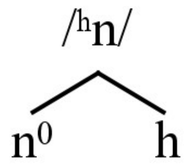
At this point, I have elucidated all possible structural configurations within Huautla Mazatec obstruents. Simple obstruents lack any subsegmental specification: (t). Aspirated, glottalised, and prenasalised obstruents exhibit a single layer with one specified subsegment: (t (h)), (t (ʔ)), and (t (n)), respectively. Pre-aspirated obstruents comprise two layers, with only the lower layer containing a specified subsegment, while the first layer is void: (t (Ø) ((h))). Finally, post-aspirated, post-glottalised, pre-aspirated, and pre-glottalised pre-nasalised obstruents encompass two layers and two specified subsegments: (t (h) ((n))), (t (ʔ) ((n))), (t (n) ((h))), and (t (n) ((ʔ))).

Simple obstruents, such as (t), also adhere to the subsegmental rules proposed for complex obstruents during their derivation. Initially, (t) fills all three subsegmental positions, resulting in (t t t). Since no dependent subsegments exist, the rule governing dependent subsegment placement is inapplicable. In the third step, a default rule assigns voicelessness, yielding ($t_{[0\text{voice}]}$ $t_{[0\text{voice}]}$ $t_{[0\text{voice}]}$) → ($t_{[\text{-voice}]}$ $t_{[\text{-voice}]}$ $t_{[\text{-voice}]}$). Finally, the rule for erasing inner parenthetical boundaries does not apply, resulting in the final representation: (t t t).

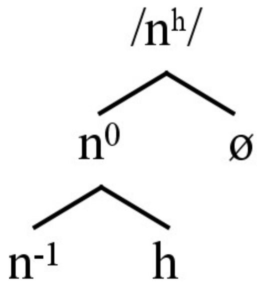
Sonorants in this system can potentially contrast as plain /n/, pre-glottalised /$^{\text{ʔ}}$n/, pre-aspirated /^{h}n/, post-glottalised /n$^{\text{ʔ}}$/, and post-aspirated /n^{h}/[17]. Plain sonorants are represented as (n). Pre-glottalised and pre-aspirated sonorants are represented as (n (ʔ)) and (n (h)), respectively (13). Post-glottalised and post-aspirated sonorants are represented as (n (Ø) ((ʔ))) and (n (Ø) ((h))), respectively (14).

(13)

[17] According to G. Starostin (personal communication, June 2025), in S. Starostin's (1994: 500) proposed reconstruction of the transitional state from Old Chinese to Classical Old Chinese (ca. 5th-2nd centuries BC), this language featured a series of 'voiceless sonorants' in contrast with a series of post-aspirated sonorants. This implies that at some point in its history, Old Chinese possessed a distinction identical to the one proposed here for Huautla Mazatec, namely a differentiation between pre- and post-aspirated sonorants.



(14)



The derivation of sonorants differs from that of obstruents in that it takes fewer rules to achieve its phonetic manifestation. Specifically, the rule specifying the voicing of the head subsegment by the influence of the dependent subsegment is not applicable. This is because sonorants are inherently voiced.

The derivation of (n (h)) involves three sequential steps: filling the required missing subsegment ((n (h)) → (n n (h))), moving the glottal subsegment to the left edge due (n) occupying the X^0 position ((n n (h)) → ((h) n n)), and finally, erasing the inner parenthetical boundaries: ((h) n n) → (h n n).

The derivation of (n (Ø) ((h))) necessitates two cycles of rule application due to its two-layered structure. In the first cycle, the missing subsegment is filled, resulting in (n n (Ø) ((h))). Subsequently, inner layer erasure occurs, eliminating the void position, yielding (n n (h)). The second rule is inapplicable in this cycle due to the absence of a specified subsegment in the first layer. The second cycle involves two rules. Firstly, the anti-KBP rule is applied, maintaining the subsegment (h) at the right edge given the sonorant in the X^0 position: (n n (h)). Finally, the inner boundaries are deleted, resulting in (n n h).

Having examined all possible subsegmental structures within Huautla Mazatec phonemes, I now present a revised version of Table 2, incorporating their subsegmental representations.

Table 3. Subsegmental representation of Huautla Mazatec Consonants

	Surface manifestation	Labial	Alveolar	Post-alveolar	Retroflex	Palatal	Velar	Glottal
Plosives	Plain	(p)	(t)				(k)	(ʔ)
	Pre-aspirated		(t (Ø)((h)))				(k (Ø)((h)))	
	Post-aspirated		(t (h))				(t (h))	
	Glottalised		(t (ʔ))				(k (ʔ))	
	Pre-nasalised	(p (n))	(t (n))				(k (n))	
	Pre-aspirated pre-nasalised		(t (n)((h)))				(k (n)((h)))	
	Pre-glottalised pre-nasalised	(p(n)((ʔ)))	(t (n)((ʔ)))				(k (n)((ʔ)))	
	Post-aspirated pre-nasalised		(t (h)((n)))				(k (h)((n)))	
	Post-glottalised pre-nasalised		(t (ʔ)((n)))				(k (ʔ)((n)))	
Affricates	Plain		(t͡s)	(t͡ʃ)	(ʈ͡ʂ)			
	Pre-aspirated		(t͡s(Ø)((h)))	(t͡ʃ(Ø)((h)))	(ʈ͡ʂ(Ø)((h)))			
	Post-aspirated		(t͡s (h))	(t͡ʃ (h))	(ʈ͡ʂ (h))			
	Glottalised		(t͡s (ʔ))	(t͡ʃ (ʔ))	(ʈ͡ʂ (ʔ))			
	Prenasalised		(t͡s (n))	(t͡ʃ (n))	(ʈ͡ʂ (n))			
	Pre-aspirated pre-nasalised		(t͡s (n)((h)))	(t͡ʃ (n)((h)))	(ʈ͡ʂ (n)((h)))			
	Pre-glottalised pre-nasalised		(t͡s (n)((ʔ)))	(t͡ʃ (n)((ʔ)))	(ʈ͡ʂ (n)((ʔ)))			
	Post-aspirated pre-nasalised		(t͡s (h)((n)))	(t͡ʃ (h)((n)))	(ʈ͡ʂ (h)((n)))			
	Post-glottalised pre-nasalised		(t͡s (ʔ)((n)))	(t͡ʃ (ʔ)((n)))	(ʈ͡ʂ (ʔ)((n)))			
Fricatives	Plain		(s)		(ʂ)			(h)
	Aspirated		(s (h))		(ʂ (h))			
	Glottalised		(s (ʔ))		(ʂ (ʔ))			
Nasals	Plain	(m)	(n)			(ɲ)		
	Pre-aspirated	(m (h))	(n (h))			(ɲ (h))		
	Post-aspirated	(m(Ø)((h)))	(n (Ø)((h)))					
	Pre-glottalised	(m (ʔ))	(n (ʔ))			(ɲ (ʔ))		
	Post-glottalised	(m(Ø)((ʔ)))	(n (Ø)((ʔ)))			(ɲ (Ø)((ʔ)))		
Laterals	Plain		(l)					
	Pre-glottalised		(l (ʔ))					
	Post-glottalised		(l (Ø)((ʔ)))					
Trills			(r)					
Taps			(ɾ)					
Approximants	Plain	(β)				(j)		
	Pre-aspirated	(β (h))				(j (h))		
	Post-aspirated	(β (Ø)((h)))				(j (Ø)((h)))		
	Pre-glottalised	(β (ʔ))				(j (ʔ))		
	Post-glottalised	(β (Ø)((ʔ)))				(j (Ø)((ʔ)))		

This new representational framework allows for a re-evaluation of the richness of the Huautla Mazatec consonant inventory. While the inventory of 'typical' phonemes, namely those lacking secondary articulations, comprises only 18 segments, these phonemes form clusters of correlations with complex consonants. These correlations arise from contrasts mediated by the action of just three possible subsegments: (n), (h), and (ʔ), which can be considered 'satellites' as per Steriade (1994)

Combinations between the 18 principal subsegments, corresponding to the 'typical' phonemes and the three possible subsegments are mediated by four distinct structural templates: (X), (X (Y)), (X (Y) ((Z))), and (X (Ø) ((Y))). The filling of these subsegmental variables is subject to constraints inherent to Huautla Mazatec. While X subsegments can be filled by any of the distinctive phonemes of the language, restrictions apply when X represents a sonorant: Z subsegments are prohibited, and (n) cannot occupy the Y subsegment position.

By applying these meta-rules, we can predict that Huautla Mazatec sonorants exhibit five distinct correlative patterns: one derived from the (X) template, two from the (X (Y)) template ((X (h)) and (X (ʔ))), and two from the (X (Ø) ((Y))) template ((X (Ø) ((h))) and (X (Ø) ((ʔ)))). This prediction is corroborated by the empirical observation of five distinct sonorant series: plain, pre-glottalised, pre-aspirated, post-glottalised, and post-aspirated.

Conversely, obstruent subsegments are subject to three combinatorial restrictions previously identified by Steriade (1994: 205): two glottal subsegments cannot co-occur within a single phoneme, the glottal stop (ʔ) cannot precede any obstruent, and the nasal (n) cannot follow any obstruent. These restrictions constrain the possible combinations, leading to the prediction of nine distinct series: one derived from the simplest structure, (X), three from (X (Y)), four from (X (Y) ((Z))), and one from (X (Ø) ((Y))). These nine series precisely correspond to the nine observed obstruent series in Huautla Mazatec:

1) Plain (X)

2) Pre-nasalised (X (n))

3) Post-aspirated plain (X (h))

4) Post-aspirated pre-nasalised (X (h)((n)))

5) Post-glottalised plain (X (ʔ))

6) Post-glottalised pre-nasalised (X (ʔ)((n)))

7)Pre-aspirated plain (X (Ø)((h)))

8) Pre-aspirated pre-nasalised (X (n)((h)))

9) Pre-glottalised pre-nasalised (X (n)((ʔ)))

Although the proposed templates might appear to be derived from a tautological assumption, two key predictions can be made to justify their validity. Firstly, as observed in Table 3, there are notable distributional gaps within the clusters of correlations formed between phonemes with the (X) template. For instance, in the labial plosive cluster, only three of the nine possible templates are attested: (X), represented by (p) (/p/); (X (Y)), represented by (p (n)) (/^{n}p/); and (X (Y) ((Z))), represented by (p (n)((ʔ))) (/$^{\text{ʔn}}$p/).

In Huautla Mazatec, as with other Eastern Otomanguean languages (Campbell 2017a, 2017b; Kaufman 2015), labial plosives represent a recent innovation acquired through contact with Spanish. For example, the phoneme /p/ appears in words like /pa^{4}/ 'father,' borrowed from the Spanish word 'papá,' and /pa^{3}ɲo^{1}/ 'shawl,' adapted from the Spanish 'paño.' Similarly, the pre-nasalised /np/ appears in the word /^{n}pa^{3}/ [^{m}ba^{3}], meaning 'son's godfather' or 'buddy,' which is an adaptation from the Spanish word 'compadre' (the source of this borrowing). Notably, the source word in Spanish contains a heterosyllabic sequence [...m.pa...], which becomes /^{n}pa/ in Huautla Mazatec.

Gap-filling within phonological systems is a well-documented phenomenon. As noted by Nikolaev (2022: 161), it frequently occurs during phoneme borrowing and is also a significant driver of diachronic phonological change (Campbell 2013: 68-71; Salmons & Honeybone 2015). This process demonstrates a tendency towards the "symmetry ideal" proposed by Trubetzkoy (1939). I propose that phonological systems tend to fill gaps by utilizing existing subsegmental templates.

The emergence of /p/ in Mazatec, likely resulting from contact with Spanish, filled a gap in the intersection of the labial and plosive series, occupying the (X) template. Subsequently, the remaining eight available templates for plosives in Huautla Mazatec could be utilised for further gap-filling. This pattern is evident in the emergence of /^{n}p/, which fills the (X (n)) template. A third labial plosive, /^{ʔn}p/, is attested in the adjective /^{ʔn}pa^{4}/ [^{ʔm}ba^{4}] 'huge.'

Campbell (2013: 68) observe that borrowed sounds, particularly those associated with onomatopoeia or expressive qualities, are frequently incorporated into the phonological system of the borrowing language. However, the presence of /^{ʔn}p/ in Huautla Mazatec presents an intriguing case. While /^{ʔn}p/ appears in the word /^{ʔn}pa^{4}/ [^{ʔm}ba^{4}], it is unlikely to have originated from Spanish, as that language lacks any phonological contrast involving glottalisation. Its most probable origin appears to be innovation driven by sound symbolism; that is, its phonological form seems to have arisen from synesthetic motivation.

I propose that /^{ʔn}p/ arose through a gap-filling process within the existing subsegmental template system. Specifically, /^{ʔn}p/ fills the (X (n) ((ʔ))) template. Importantly, the emergence of /ˀ^{n}p/ was contingent upon the prior existence of /^{n}p/ within the phonological system. This suggests that the (X (n)) template, which underlies /^{n}p/, served as the foundation for the subsequent development of the more complex (X (n) ((ʔ))) template and the corresponding /ˀ^{n}p/ phoneme.

This is not an isolated instance of gap-filling within the subsegmental framework of Huautla Mazatec. The word /lˀi^{4}/ 'fire' exemplifies this phenomenon. While /lˀi^{4}/ represents a regular development from Proto-Mazatec (*cf.* Chávez Peón et al. 2025; Kirk 1966), its post-glottalised lateral constituent is noteworthy, as it contradicts the general tendency for glottalisation to precede sonorants, as predicted by KBP.

However, there exists a word, /le^{2}ˀle^{1}/ 'listless,' which exhibits the expected pre-glottalised lateral. Notably, it seems that /le^{2}ˀle^{1}/ is also sound-symbolic word and cannot be reconstructed from Proto-Mazatec as it is absent in other Mazatec languages. This observation suggests that the presence of /lˀ/, which corresponds to the (X (Ø) ((ʔ))) template, may have facilitated the gap-filling of the simpler (X (ʔ)) template.

A second key prediction of this framework allows us to posit a markedness scale among subsegmental templates, directly correlate with their expected frequency within the lexicon. I propose two potential factors that contribute to template markedness: structural elaboration and the number of present subsegments.

To evaluate the relative importance of these factors, Table 5 presents the results of a quantitative analysis of template frequencies. This analysis utilised the 'Vocabulario Mazateco' of Eunice Pike (1957) as the corpus. This corpus comprises a total of 5511 individual phoneme occurrences within 1108 Huautla Mazatec lexemes, encompassing simple words, compound words, and idioms[18]. I will compare these results with general typological tendencies using the PHOIBLE database 2.0 (Moran & McCloy 2019).

Table 4. Quantification of Huautla Mazatec consonant phonemes' templates present in Pike (1957).

	(X)	(X (Y))	(X (Y) ((Z)))	(X (Ø) ((Y)))	Total
Number of occurrences	3311	1477	332	391	5511
Percentage	60.07%	26.80%	6.02%	7.09%	100%

The results presented in Table 4 reveal a clear preference for the (X) template in Huautla Mazatec. This observation is unsurprising, as all spoken human languages include plain phonemes, a fact supported by the PHOIBLE database (Moran & McCloy 2019), which reports a 100% occurrence rate of plain phonemes across its sampled languages.

The (X (Y)) template, representing complex phonemes with one secondary articulation in their surface form, occurs in slightly more than a quarter of the analysed corpus of Huautla Mazatec. The lower frequency of (X (Y)) consonants compared to plain (X) ones in Huautla Mazatec is expected assuming two parameters mentioned above, as they count with more than one subsegment and bigger structural elaboration, and no void layers.

Although the PHOIBLE database does not provide an exact number on the percentage of languages that have consonantal phonemes with one secondary articulation in their surface manifestation, the most common contrastive consonantal phonemes with a secondary articulation in their phonetic form are k^h and p^h, both present in 20% of the sample. If we extrapolate this number to the assumption that the number of languages with the (X (Y))

[18] The analysed words were exclusively those listed in the vocabulary entries found within pages 1 to 70 of the alphabetically listed vocabulary section. All other words appearing in different sections of the text were excluded from the analysis. Words listed as initiating with vowels were considered instances of (X), assuming an initial glottal stop (/ʔ/), while instances of letter f are considered instances of (X (Y)), assuming the presence of /^{h}β/. The quantification of these occurrences was conducted manually using the Multi Counter App for Android.

template is around that figure, it seems that its distribution is similar, albeit a little lower, when compared to the distribution of these consonants in Huautla Mazatec.

The similar lower distribution of (X (Ø) ((Y))) and (X (Y) ((Z))) is expected, as they both exhibit greater structural complexity, that is, more than one layer. Although (X (Y) ((Z))) has a higher number of subsegments, and thus might reflect a lower frequency compared to (X (Ø) ((Y))) as shown in Table 4, this difference is negligible without a more detailed study that considers the selection of analysed items and incorporates statistical weighting to the results.

Regarding the typological distribution of these templates, we are now entering uncharted territory. Since these templates are a consequence of our current analytical proposal and are justified by the establishment of rules that govern their operation, we would need to delve into the inner workings of the phonological system of each language on a case-by-case basis to justify the proposal of such templates.

However, we can attempt to approach the existence of these templates in other languages. For the (X (Ø) ((Y))) template, I have detected two potential cases in other Otomanguean languages: Valley Zapotec and Quiotepec Chinantec. Although I have described these templates thus far only with reference to consonants, they can also be used to describe complex vowels, such as nasal, non-modal, or pharyngealised vowels. For example, the four nasal vowels of Huautla Mazatec can be represented as (i (n)), (e (n)), (o (n)), and (a (n)).

Valley Zapotec languages like Güilá (Arellanes Arellanes 2009) and Quiaviní (Chávez Peón 2010), have been described as having a typologically unexpected contrast between two degrees of laryngealisation in vowels. Although the exact surface expression of these laryngealisations varies depending on other factors, primarily tone, the consistent distinction between them is that one degree of laryngealisation is always stronger than the other within the same context.

I propose that vowels specified with the strong degree of laryngealisation can be represented as (V (ʔ)), while those specified with the weaker degree adopt the (X (Ø) ((Y))) template: (V (Ø) ((ʔ))). In this case, I predict that the rule that determines the implementation of the subsegment (ʔ) will have a different outcome depending on its position. If (ʔ) is in the first cycle (V (ʔ)) → V̰̰, an optimal laryngeal surface will be achieved. However, if (ʔ) is in the (V (Ø) ((ʔ))) template, its implementation occurs in the second of two cycles: (V (Ø) ((ʔ))) → (V (ʔ)) → V̰, resulting in a sub-optimal laryngeal surface.

This solution is identical to the one I propose for describing the two degrees of contrastive vowel nasality in Quiotepec Chinantec (Castellanos Cruz 2014; Chávez Peón Herrero et al. 2014)[19], where vowels with the strong nasal degree are represented as (V (n)) and those with the lower degree as (V (Ø) ((n))). The optimal and suboptimal implementations depend on the cycle in which they are applied.

Although the proposed processes for deriving (V (Ø) ((n))) templates differ between Huautla Mazatec and Valley Zapotec and Quiotepec Chinantec – the former relying on the application of two different rules (KBP vs. anti-KBP), while the latter relies on different strength outcomes within the same rule – they share the crucial characteristic that the different effects depend on the layer at which the rules are applied.

At the same time, I predict that the effects of rules applied in the first layer will always be more preferred from an acoustic point of view than the effects of rules applied in successive layers. In Huautla Mazatec, post-aspiration of obstruents is preferred over pre-

[19] This contrast has been also proposed for Palantla Chinantec (Blevins 2004; Merrifield 1963).

aspiration, and the inverse is true for sonorants (Kingston 1985). Similarly, a stronger degree of nasality or laryngealisation will be preferred over a weaker degree in Quiotepec Chinantec, and Valley Zapotec, respectively.

The presence of the (X (Y) ((Z))) template in other languages can be confirmed by the existence of prenasalised post-aspirated phonemes that contrasts with both plain, prenasalised, and post-aspirated obstruents, for example, Mazahua (Knapp Ring 2008), Xhosa (Gowlett 2003), or Soghpo Tibetan (Suzuki 2011) (Cf. Moran & McCloy 2019).

Although consonants with two or more secondary articulations are attested, albeit rarely, in other languages, we cannot determine the internal constituency of their templates without a prior analysis of their phonological relations. These consonants tend to appear in systems with large inventories, arising from the intersection of correlations among series of consonants with one secondary articulation. Examples include Lezgian (Haspelmath 1993), Hakka (Lee & Zee 2009), and Huautla Mazatec. The most frequent phonemes with more than one secondary articulation found in the PHOIBLE database are k^{w}’ and k^{wh}, occurring in 2% and 3% of the languages, respectively. These figures suggest that the existence of such phonemes is highly disfavoured in the conformation of phonemic systems.

A final prediction I can propose, and potentially prove, is that phonemes with more elaborate templates are more prone to be lost during the phonological diachronic development of languages that have them. This often involves simplifying their structural elaboration by erasing weaker elements, namely, void and lower layers. This simplification can then cause these phonemes to merge with the corresponding series of phonemes with simpler templates.

As Chávez Peón & Filio (2021: 39-40) report, some speakers of Huautla Mazatec are currently losing the distinction between pre-aspirated and pre-glottalised versus post-aspirated and post-glottalised sonorants, merging them in favour of the former series, as expected under the Kingston Binding Principle. Within this model, we can propose that when /n^{h}/ neutralises to /^{h}n/, the change in its representation is a simple process of deleting the inner parenthetical boundaries and the void layer: (n (Ø)((h))) → (n (h)).

Another example from Puebla Mazatec will be provided as evidence. While conservative speakers of Puebla Mazatec have pre-aspirated plosive phonemes (e.g., /^{h}t/), innovative speakers are merging them with the corresponding plain plosive series (e.g., /t/) (Carrera Guerrero 2025; Wagner Oviedo 2017). This process involves the deletion of all the inner elements except the head subsegment (t): (t (Ø)((h))) → (t).

Further investigation of other languages could bring to light the existence of other types of templates. For example, Ixcatlán Mazatec has post-glottalised prenasalised palatalised obstruents, such as the one found in the word [^{ŋ}k^{ʔj}o^{3}] ‘cocoa’. Wagner Oviedo (2018) proposes a strict layering order of subsegments: place subsegments (j, w) > glottal subsegments (h/ʔ) > nasal subsegment (n)[20]. The representation of [^{ŋ}k^{ʔj}] would then be (k (j)((ʔ))(((n)))), an example of an (X (Y)((Z)))(((Z'))) template.

I should highlight that the concept of complex phoneme also includes doubly articulated segments, such as affricates or plosives with two places of articulation in their surface manifestation. A complex phoneme’s subsegmental representation differs from that of consonants that include secondary articulation in their surface manifestation by lacking inner layers (Cf. Garvin et al. 2018; Inkelas & Shih 2016; Shih & Inkelas 2019; van de Weijer 1996, 2011), that is, (X Y).

[20] Ixcatlán Mazatec does not have pre-aspirated obstruents nor post-aspirated prenasalized consonants (Mario Ernesto Chávez Peón Herrero & Filio García 2021; Wagner Oviedo 2018).

For example, the contrast in Saramaccan between /k/, /k^w/, and /k͡p/ (Aboh et al. 2013) would be represented as (k), (k (w)), and (k p), respectively. It is even possible to encounter complex phonemes with both a doubly articulated and a secondary articulation simultaneously, such as the phoneme /$^{\widehat{ŋm}}$g͡b/ found in several languages across Central Africa (Moran & McCloy 2019). Its subsegmental representation would follow the template (X Y (Z)), that is, (g b (n)).

In this section, I have presented a novel approach to phonologizing complex phonetic sequences in Huautla Mazatec. This approach adapts P&P (1947) concept of 'Immediate Constituents' to the subsegmental representation framework within Q-theory (Garvin et al. 2018; Shih & Inkelas 2019), while also incorporating binary hierarchical structures with head and dependent positions to represent the underlying forms of Huautla Mazatec phonemes. These binary structures allow us to represent the properties of projection of head subsegments into lower layers (X^1 (…X^n)), as well as the possibility of the occurrence of void layers (Ø) within dependent layers

A further meta-analysis of the representations of these structures leads to the proposal of subsegmental templates: abstract structures that can be filled with variable subsegmental values. I have shown that the analysis of these templates can allow us to predict further characteristics related to the phonemic inventory of Huautla Mazatec, such as the incidence of each type of template within the lexicon, or to explain the phenomena of gap-filling within the historical development of phonemic systems.

4. Conclusions

In this paper, I have argued for the existence of a subsegmental structure within phonemes to explain the patterns observed in Huautla Mazatec consonantal phonemes, as well as the rules that mediate between underlying and surface representations. By proposing a combination of 18 subsegments (corresponding to the 18 non-complex phonemes of Huautla Mazatec), 3 'satellite' subsegments ((n), (h), and (ʔ)), and four subsegmental templates ((X), (X (Y)), (X (Ø) ((Y))), and (X (Y)((Z)))), we can account for the large inventory of 85 contrastive phonemes in this language, while adhering to the monophonematic rules proposed by Trubetzkoy (1939).

By employing Q-theory representations (Garvin et al., 2018; Inkelas & Shih, 2016; Shih & Inkelas, 2019) within P&P's (1947) 'Immediate Constituents', and utilizing the concept of subsegments, we can propose a structured framework suitable to address the question how much phonetics should be considered within phonology. I propose that, while subsegments and their phonetic specifications are primarily specified underlyingly, the temporal sequencing of these subsegments, particularly in the case of surface complex phonemes, is determined through derivation.

I acknowledge that this discussion has not addressed the integration of phonological features within subsegments. However, the abstract nature of the subsegments and templates allows for considerable flexibility in their instantiation. This flexibility would enable the seamless incorporation of various feature models or phonological primes. These could include, for example, Feature Geometry (Clements 1985; Hall 2007), Particle Phonology (Schane 1984; van de Weijer 1996), Element Theory (Backley 2011; Nasukawa 2017, this Volume), Substance-Free or Emergent Features-based approaches (Mielke 2008; Scheer 2022). The only crucial assumption is that the features assigned to the dependent position of each layer should correspond to articulatory gestures that are distinct from the major features (i.e., place, manner, and voice for consonants; and height, backness, and rounding for vowels). Alternatively, it could accommodate even 'seglets' ordered in a sub-phonemic hierarchical structure integrated into the prosodic structure of the language, following Golston and Kehrein's proposal in this volume. This would involve adjusting the theoretical status of manner in consonants and rounding in vowels to the particular requirements of that model.

I must remark that the existence of branching structures within phonology has been proposed by Anderson (1986) under the concept of 'Structural Analogy'. It has also been observed by den Dikken & van der Hulst (2020) in their exploration of analogies between syntax and phonology. As van der Hulst (this volume) remarks, 'the 'One Syntax for All' program assumes that phonology and syntax have recourse to the same computational system, which includes structural notions such as hierarchical structure, binarity, headedness, and self-embedding recursion.' This paper contributes to this discussion by proposing the existence of binary structures at the subsegmental level of the phonological segment, thereby increasing the evidence for structural analogies between phonology and syntax.

Finally, evaluating Huautla Mazatec subsegmental template patterns in light of the typological tendencies of phonological systems allowed us to demonstrate the broader validity of this proposal beyond the specific case of Huautla Mazatec. Furthermore, this framework could potentially be applied to explain other types of phenomena in other languages related to subsegmental complexity.

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